

FOGO SMART FRAME

Passive Airflow-Based Anti-Fog Spectacle Frame Concept

Project Documentation & Visual Concept Showcase

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Hackathon: AI Conclave 2026 – Hackathon

Organizer: Amal Jyothi College of Engineering

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Figure 1. Fogo Smart Frame concept visualization.

Clear Vision, Everyday Comfort.

1. Project Overview

Fogo Smart Frame is a conceptual anti-fog spectacle frame designed to explore a passive way of reducing lens fogging caused by warm, moisture-rich breath. The concept uses airflow guidance around the nose bridge and upper frame rather than depending on batteries, motors, or continuous electronic control.

The design is intended as a lightweight and discreet concept that can be explored for prescription eyewear and protective-use situations.

2. Problem Statement

Warm humid breath can reach a cooler lens surface and contribute to condensation and fogging. The issue can become more noticeable when masks or protective equipment are worn, during outdoor riding, around warm steam, or when moving between different temperature environments.

Potential situations include:

- Mask usage and close-to-face breathing
- Healthcare and laboratory work
- Construction and protective-wear environments
- Bike riding and outdoor activities
- Exposure to warm food or steam
- Movement between warm and air-conditioned spaces

3. Proposed Solution

The Fogo concept combines a thin transparent silicone airflow guide near the nose bridge with small airflow channels along the upper frame. The intended airflow path redirects warm breath away from the lens area and toward side exits.

The core version is passive: the geometry of the physical structure guides airflow without requiring a battery or electronic prediction system.

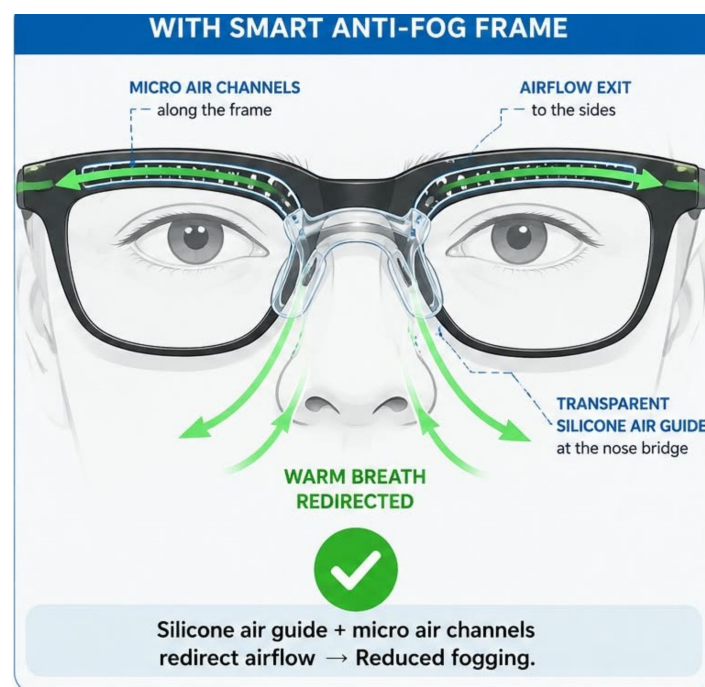


Figure 2. Close-up concept showing the proposed airflow path and components.

4. How It Works

1. **Warm breath is generated:** Exhaled air contains heat and moisture.
2. **Airflow is captured/guided:** The nose-bridge guide helps direct airflow away from the lens surface.
3. **Air moves through the frame channels:** The proposed channels provide a guided path toward the sides.
4. **Air exits laterally:** Side openings allow redirected airflow to leave the frame area.
5. **Fogging may be reduced:** Lower direct exposure of the lens to warm humid airflow can help reduce condensation under suitable conditions.

Important: The illustrations show the intended airflow concept; they are not measurements or proof of a particular airflow rate or fog-reduction percentage.

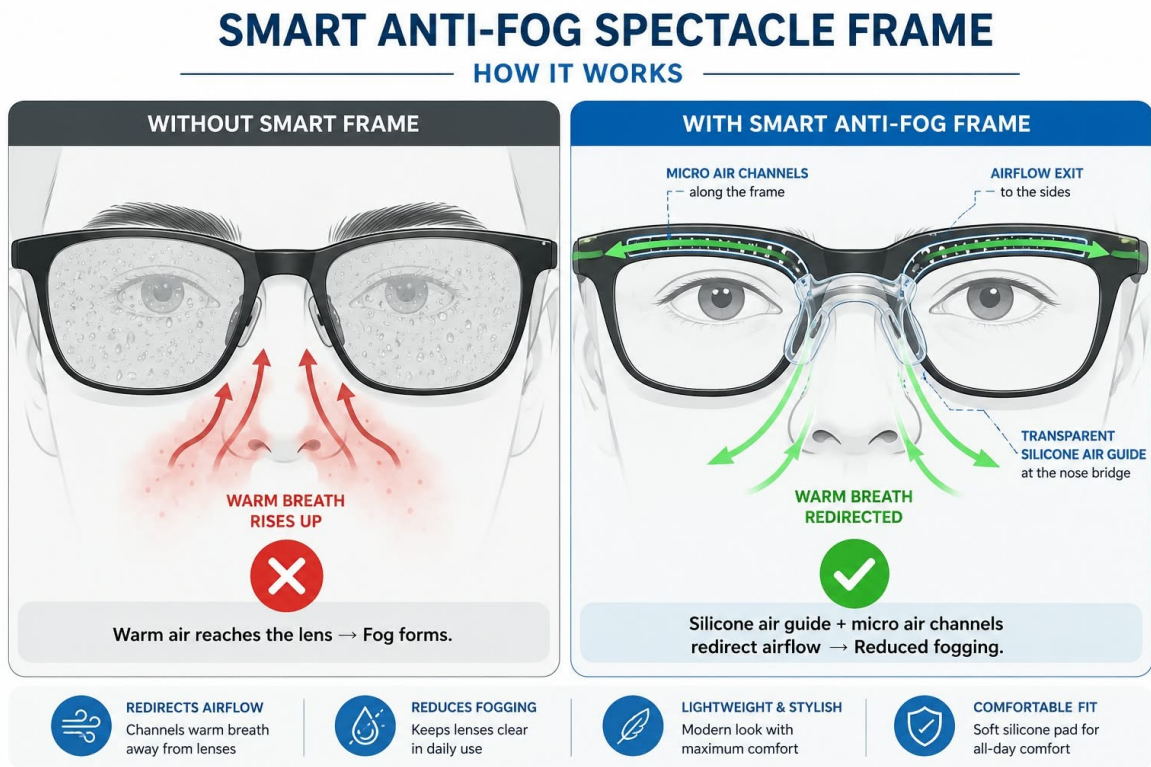


Figure 3. Concept comparison: conventional airflow versus the proposed guided-airflow design.

5. Main Design Features

Feature	Conceptual purpose
Transparent silicone air guide	Provides a discreet route for redirecting warm breath near the nose bridge.
Micro airflow channels	Guide airflow toward the sides of the frame.
Passive operation	Designed without batteries, motors, or active electronics in the core version.
Lightweight structure	Aims to support everyday wearing comfort.
Prescription compatibility	Can be explored for integration with ordinary prescription eyewear.
Reusable concept	Components/materials can be developed for repeated use and cleaning.

6. Applications

The concept is intended to be explored in situations where users experience recurring eyewear fogging. The following visual presents potential application contexts.



Figure 4. Illustrative application scenarios for the concept.

Potential user groups include healthcare workers, laboratory workers, construction workers, bike riders, students, office users, and people who frequently wear masks or protective equipment.

7. Design Philosophy

Simple • Lightweight • Reusable • Battery-free • Practical

The core design prioritizes physical airflow management instead of making the solution dependent on electronics. This can potentially simplify maintenance and reduce the number of components.

8. AI / Future Technology Possibilities

AI is not required for the core passive Fogo concept. A future advanced version could explore temperature/humidity sensors and AI-assisted control to monitor environmental conditions and support adaptive airflow management. Such an electronic version would be a future direction, not part of the core passive design presented here.

9. Development & Testing Plan

Future development can include:

- Fabricating a physical prototype and checking fit and comfort
- Testing different channel sizes and opening geometries
- Comparing fogging behavior under controlled temperature and humidity conditions
- Studying airflow with simple visualization or computational simulation
- Testing cleaning, durability, and repeated-use behavior
- Exploring compatibility with different prescription frame geometries

10. Limitations

This project is currently a concept/prototype-stage design. Actual performance will depend on face fit, frame geometry, breathing pattern, humidity, temperature, wind, materials, and the final manufacturing process. Controlled experimental validation is required before making specific performance claims or considering safety-critical or medical use.

11. Hackathon Participation

AI Conclave 2026 – Hackathon

Organized by Amal Jyothi College of Engineering

Held on 16 September 2026

Fogo Smart Frame was developed and presented as a project concept during the hackathon.

12. Repository Contents

The GitHub repository can contain the README, project images, demonstration video, and this documentation PDF.

13. Conclusion

Fogo Smart Frame explores a practical physical approach to an everyday eyewear problem: fogging caused by warm, humid airflow. By guiding airflow through a thin frame structure, the concept aims to reduce direct exposure of the lens to exhaled air without requiring a battery-powered system.

Further prototyping and controlled testing are needed to validate the airflow design, comfort, durability, manufacturability, and real-world anti-fog performance.

Fogo Smart Frame — Clear Vision, Everyday Comfort.

Visuals in this document are concept illustrations prepared for project communication. They should not be interpreted as measured engineering results or product certification.